

AMENDMENT NO. 1 APRIL 1995
TO
IS 3195 : 1992 STEEL FOR THE MANUFACTURE OF
VOLUTE AND HELICAL SPRINGS (FOR RAILWAY
ROLLING STOCK) — SPECIFICATION
(Third Revision)

(Second cover page, Foreword, line 7) – Substitute ‘51CrMoV’ for ‘51CrMoV4’.
(Page 2, Table 1) – Substitute the following for the existing table:

Table 1 Chemical Composition
(Clauses 3, 6.1 and 6.2)

Grade	Designation [see IS 1762 (Part 1) : 1974]	Constituents, Percent							
		Carbon	Silicon	Manganese	Sulphur Max	Phosphorus Max	Chromium	Vanadium	Molybdenum
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1	98C6	0.90 to 1.05	0.15 to 0.35	0.50 to 0.80	0.050	0.050	—	—	—
2	113C6	1.05 to 1.20	0.15 to 0.35	0.50 to 0.80	0.050	0.050	—	—	—
3	55Si7	0.50 to 0.60	1.50 to 2.00	0.80 to 1.00	0.030	0.030	—	—	—
4	60Si7	0.55 to 0.65	1.50 to 2.00	0.80 to 1.00	0.030	0.030	—	—	—
5	65Si7	0.60 to 0.70	1.50 to 2.00	0.80 to 1.00	0.030	0.030	—	—	—
6	50Cr4V2	0.45 to 0.55	0.15 to 0.35	0.50 to 0.80	0.030	0.030	0.90 to 1.20	0.15 to 0.30	—
7	60Cr4V2	0.55 to 0.65	0.15 to 0.35	0.80 to 1.00	0.030	0.030	0.90 to 1.20	0.15 to 0.30	—
8	51CrMoV	0.48 to 0.54	0.15 to 0.40	0.70 to 1.10	0.030	0.030	0.90 to 1.20	0.07 to 0.12	0.15 to 0.25

NOTE – The limiting diameters for bars for the manufacture of springs are indicated below for guidance.

- a) 55Si7 : Recommended for bars up to 32 mm dia
- b) 60Si7 : Recommended for bars up to 32 mm dia
- c) 65Si7 : Recommended for bars up to 32 mm dia
- d) 50Cr4V2 : Recommended for bars up to 32 mm dia
- e) 60Cr4V2 : Recommended for bars up to 47.5 mm dia
- f) 51CrMoV : Recommended for bars up to 47.5 mm dia

(Page 2, Table 2, headed under Grade) – Substitute ‘51CrMoV’ for ‘51CrMoV4’.
(Page 3, clause 8.6.1, line 3) – Substitute ‘plain’ for ‘plane’.
(Page 3, clause 8.6.2, line 2) – Substitute ‘60Cr4V2, 51CrMoV’ for ‘60Cr4V3, 51CrMoV4’.
(Page 3, clause 9, line 2) – Substitute ‘2.0’ for ‘2.5’.
(Page 3, clause 10.2, line 1; and clause 10.3, line 6) – Substitute ‘peeled’ for ‘pealed’.
(Page 3, clause 11.4 (b), line 1) – Substitute ‘Peeled’ for ‘Pealed’.

(MTD 4)

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